

MUTHU RATHNASHRI C – INFORMATION TECHNOLOGY

SOFTWARE DEVELOPMENT ENGINEER IN AMAZON

About the Company:

Amazon, a global leader in e-commerce and cloud computing, is known for its customer-centric approach and innovative technologies. Central to its "Day 1 culture" is a commitment to constant innovation, agility, and a focus on customers. As a software development engineer at Amazon, you'll work in a dynamic environment that values creativity and technical excellence. The company emphasizes scalable solutions and robust architecture, impacting millions of customers globally.

About the Role:

Software Development Engineers (SDEs intern cum placement) at Amazon play a vital role in shaping the company's technology. They design, develop, and maintain scalable software solutions that drive Amazon's services, including e-commerce and cloud computing. Their responsibilities include optimizing backend systems, enhancing data processing, and developing innovative features for applications. By leveraging advanced technologies and collaborating with a diverse team, SDEs contribute to Amazon's innovation and operational excellence, ensuring exceptional customer experiences and maintaining industry leadership.

Overview of the Process:

The overall process includes

- An online coding contest in Hackerrank platform
- Interview Round (Technical) – online mode (Amazon chime)

Online coding contest in Hackerrank platform:

During my online test, which was conducted on the HackerRank platform, I encountered two coding questions (1 easy level & 1 medium

level)focused on algorithmic problem-solving involves array manipulation and pattern matching.

Question 1:

Given an array of product costs, the task is to find the number of pairs (i,j) where $i < j$ and the sum of the costs is divisible by a given number x.

Example:

Input: x=10, costs = [3, 7, 27, 23]

Output: 4

Explanation: (3,7), (27,3), (7,23), (27,23) these 4 pairs divisible by 10

Question 2:

Given an array of regex strings containing lowercase letters and wildcard characters ('?'), the task is to find a common pattern with the minimum number of '?' characters. This pattern must intersect with all the given strings, meaning the '?' characters can be replaced to form a matching string across all patterns.

Example:

Input: n = 3 , Patterns[] = {"bug" , "gub" , "g?a"}

Output: 2

Explanation: Choose "?u?" with 2 question marks.

I have solved both questions. After completing the coding test, they conducted HR-type multiple-choice questions. Make sure to carefully read and answer each one thoughtfully. Nearly 86 students got selected to next round technical interview.

Interview Round (Technical):

This round was purely based on data structures and algorithm. In my panel there are 2 members one for asking questions and another person for observing my performance during interview.

- First, they both introduce about themselves and ask me tell me about yourself
- Then they directly jump into DSA questions since this interview process help in online mode through amazon chime they have also shared me link of liveCode amazon for code during interview.
- They asked me two DSA questions and carefully analyzed my approach to solving the problems, focusing on my choice of data structures and the complexity of my code. They also observed how I explained my solution during the dry run, specifically inquiring about time and space complexity, and asked me to justify my reasoning.
 - 1) Longest substring without repeating characters.
 - 2) Boundary traversal of binary tree.
- Before starting to code, I shared my intuition and explained the approach I planned to take. After discussing my strategy, I began coding. Once I completed the solution, the interviewer asked me to perform a dry run of the code to demonstrate its correctness.
- At the end of the interview, I was given the opportunity to ask questions. inquired about the nature of the work I would be involved in if selected, particularly focusing on the responsibilities and tasks at the early stages of an internship.

Tips and tricks that I followed:

- When I came to know that I had been shortlisted for the next round, I began researching previous interview experiences on Glassdoor and GFG archives. This gave me an idea of the areas they focus on.
- Additionally, in the pre-placement talk, they mentioned that they would primarily assess knowledge of data structures and algorithms. Amazon typically emphasizes concepts related to Trees and Graphs.
- Try to solve problems by effectively using dry runs to demonstrate your problem-solving skills and approach.
- Clearly explain both the brute-force and optimized solutions, then implement the optimized one. Ensure you have a solid understanding of the time and space complexity of your program.

My preparation strategy:

- During my two-month leave in the 6th semester, I dedicated myself to preparing for data structures and algorithms (DSA).
- I solved daily problems on LeetCode and GFG and participated in virtual contests to test my skills.
- I used materials from our seniors for CS fundamentals, ACM CODE, and Striver's AtoZ and SDE sheets to strengthen my understanding.
- In addition to DSA, I made sure to spend at least an hour each day on a mini project I started during this preparation period.
- On alternate days, I practiced aptitude questions from IndiaBix to keep my problem-solving skills sharp. Throughout this time, I focused on staying disciplined and making the most of every moment, avoiding distractions and unnecessary tasks.

- I have spent 1 weeks to study entire portion of our core subjects like OS , OOPS, CN, SQL queries, DBMS and DSA.
- I made it a habit to take notes on every problem I solved, recording the question, the brute-force approach, and the better and optimal methods to solve it. This practice was extremely helpful for quick revision, especially in the final moments before an interview.
- First, focus on solving all easy to medium problems within 45 minutes. Once you feel confident, start tackling the harder problems.
- I prefer using Overleaf for creating resumes due to its excellent formatting and spacing options. A well-formatted resume is crucial for making a professional impression and effectively showcasing your skills and experience.